

Cluster Analysis of Alzheimer’s Disease Patients

Using Brain MRI Features and Clinical Data

OASIS-1 Dataset — $n = 216$ complete-case patients

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1 Introduction

This report presents a cluster analysis of Alzheimer’s disease patients from the OASIS-1 cross-sectional dataset. The goal is to discover data-driven patient subgroups that may correspond to clinically meaningful disease stages, by combining brain MRI image features with clinical assessments.

Four experimental configurations are compared:

1. **All 5 images + clinical data** — the full feature set
2. **Clinical data only** — baseline without any imaging
3. **Images only** — imaging features with no clinical signal
4. **Best image pair + clinical** (coronal + sagittal) — a reduced but clean image set

All configurations use k -medoids clustering (FasterPAM algorithm). Two values of k are evaluated ($k = 2$ and $k = 3$) alongside a data-driven k -selection analysis.

2 Dataset and Preprocessing

2.1 OASIS-1 Dataset

The Open Access Series of Imaging Studies (OASIS-1) is a cross-sectional collection of 416 subjects aged 18–96, including cognitively normal individuals and patients with early Alzheimer’s disease. The Clinical Dementia Rating (CDR) scale is used as the ground-truth label: CDR = 0 (healthy), 0.5 (very mild dementia), 1 (mild dementia), 2 (moderate dementia).

2.2 Patient Selection

Of the 416 subjects, **216 patients** have complete records for all clinical variables (Education, SES, MMSE) and are retained for this analysis. All 216 patients have a valid CDR score.

2.3 Clinical Features

Eight clinical features are used:

Table 1: Clinical features used in experiments.

Feature	Type	Scale	Distance block
Age	Continuous	years	p1 (Euclidean)
MMSE	Continuous	0–30	p1 (Euclidean)
eTIV	Continuous	mm ³	p1 (Euclidean)
nWBV	Continuous	0–1	p1 (Euclidean)
ASF	Continuous	ratio	p1 (Euclidean)
M/F	Nominal binary	M/F	p3 (Hamming)
Education	Ordinal	1–5	p3 (Hamming)
SES	Ordinal	1–5	p3 (Hamming)

2.4 Image Features — HOG Pipeline

Each patient has up to five MRI image types: `cor` (coronal), `gfc_sag` (sagittal), `gfc_tra` (transverse), `masked_tra` (skull-stripped transverse), `sbj_sag` (native-space sagittal).

Images are preprocessed with:

- Zero-padding to square using the longest dimension, centred
- Resize to 128×128 pixels
- Histogram of Oriented Gradients (HOG): 9 orientations, 8×8 px/cell, 2×2 cells/block
 $\Rightarrow$ **8100 features** per image

2.5 Per-Image PCA

HOG features are reduced per image type using PCA. A unified variance threshold of 59.2% (the minimum across all image types at 110 components) is applied, yielding **110 PCA components per image type**.

2.6 Distance Metric — RelMS

Mixed-type features require a distance metric that handles both continuous and categorical blocks separately. The *Related Metric Scaling* (RelMS) method (`robust-mixed-dist` v0.1.26) is used with the following block layout:

$$\mathbf{X} = \underbrace{[\text{image PCA}_1 | \dots | \text{image PCA}_n]}_{\text{p1: Euclidean}} \mid \underbrace{[\text{Age, MMSE, eTIV, nWBV, ASF}]}_{\text{p1: Euclidean}} \mid \underbrace{[\text{M/F, Educ, SES}]}_{\text{p3: Hamming}}$$

3 Optimal k Selection

Three standard criteria are applied to determine the best number of clusters, evaluated on both the full feature set and clinical-only features, for $k = 2, \dots, 8$.

Table 2: k-selection results: silhouette score and Davies-Bouldin index for all 5 images + clinical and clinical-only configurations. Best value per criterion highlighted.

k	All 5 images + clinical		Clinical only	
	Silhouette $\uparrow$	DB $\downarrow$	Silhouette $\uparrow$	DB $\downarrow$
2	0.0262	3.51	0.0292	3.21
3	0.0091	6.37	0.0130	5.45
4	0.0079	6.62	0.0125	5.52
5	0.0052	6.11	0.0095	5.24
6	0.0056	5.80	0.0081	4.76
7	0.0055	5.39	0.0074	5.10
8	0.0041	5.54	0.0078	4.75

Both criteria unanimously recommend $k = 2$ for all tested configurations. Figure 1 shows the elbow, silhouette, and Davies-Bouldin curves.

$k = 2$ is statistically optimal. $k = 3$ is also reported throughout this analysis because three subgroups carry more clinical interpretability for Alzheimer’s disease staging (healthy / very mild / mild–moderate).

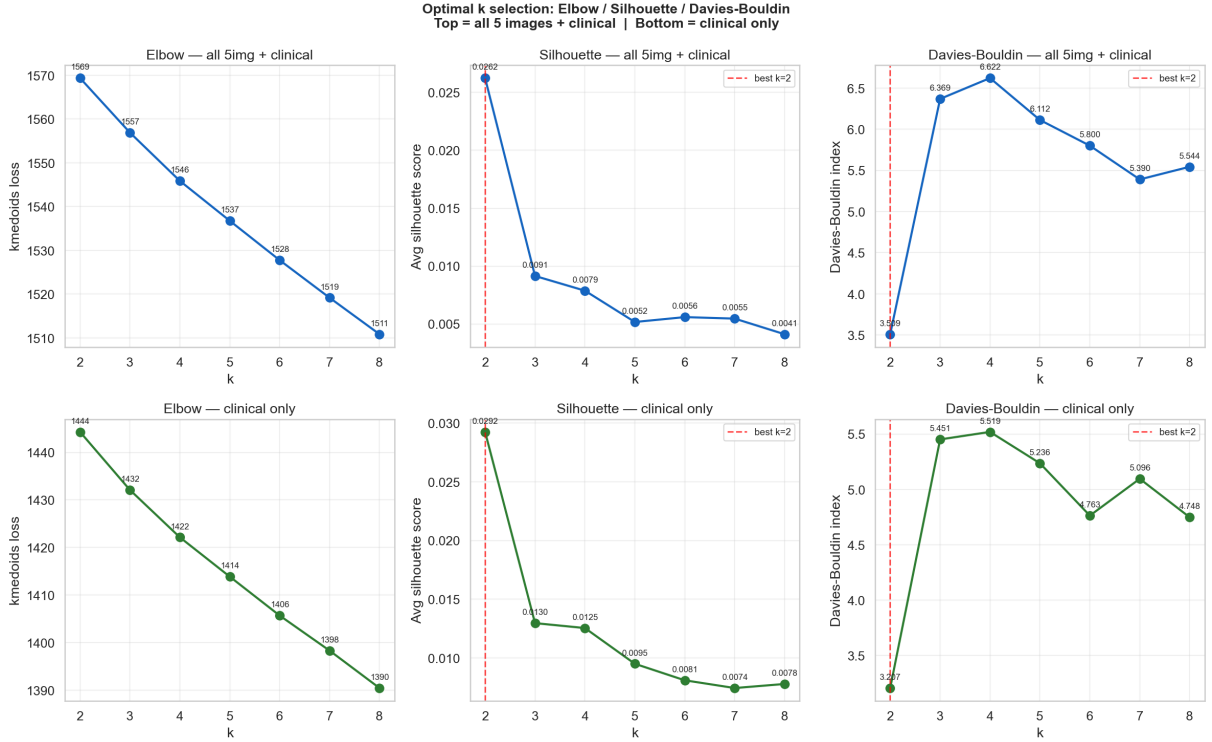


Figure 1: k-selection criteria ($k=2$ to 8). Top row: all 5 images + clinical. Bottom row: clinical only. All three methods agree on $k = 2$.

4 Experiment 1: All 5 Images + Clinical Data

4.1 Configuration

Feature matrix: five image PCA blocks (110 components each) concatenated with 5 continuous clinical features (p1) and 3 ordinal/nominal features (p3). Shape: 216×558 . Distance: RelMS.

4.2 Evaluation

Table 3: Evaluation metrics — all 5 images + clinical.

k	Silhouette	ARI vs CDR	NMI vs CDR	Sizes
2	0.0262	0.0283	0.0107	146 / 70
3	0.0091	0.0176	0.0188	65 / 56 / 95

4.3 Cluster Profiles

$k = 2$ — The dominant split is by sex and brain volume (eTIV):

Table 4: Cluster profiles — all 5 images + clinical, $k = 2$.

Feature	C0 ($n = 146$)	C1 ($n = 70$)
% Female	88.4%	22.9%
Age	72.6 ± 11.6	72.1 ± 13.8
MMSE	27.4 ± 3.5	27.2 ± 3.4
eTIV (mm^3)	1375 ± 93	1633 ± 128
nWBV	0.76 ± 0.05	0.74 ± 0.05
Education	2.95 ± 1.22	3.81 ± 1.38
SES	2.69 ± 1.03	2.07 ± 1.20
CDR=0	65.1%	54.3%
CDR=0.5	22.6%	34.3%
CDR=1	11.6%	10.0%
CDR=2	0.7%	1.4%

$k = 3$ — The split refines into three groups primarily by sex, eTIV, and education/SES:

Table 5: Cluster profiles — all 5 images + clinical, $k = 3$.

Feature	C0 ($n = 65$)	C1 ($n = 56$)	C2 ($n = 95$)
% Female	76.9%	23.2%	86.3%
Age	71.3 ± 11.7	72.4 ± 13.5	73.2 ± 12.1
MMSE	26.8 ± 3.8	27.6 ± 3.0	27.5 ± 3.4
eTIV	1430 ± 96	1661 ± 123	1359 ± 95
Education	2.28 ± 1.10	4.18 ± 1.15	3.33 ± 1.14
SES	3.29 ± 1.06	1.71 ± 0.95	2.40 ± 0.88
CDR=0	56.9%	57.1%	67.4%
CDR=0.5	23.1%	32.1%	25.3%
CDR=1	18.5%	10.7%	6.3%
CDR=2	1.5%	0.0%	1.1%

4.4 Visualisations

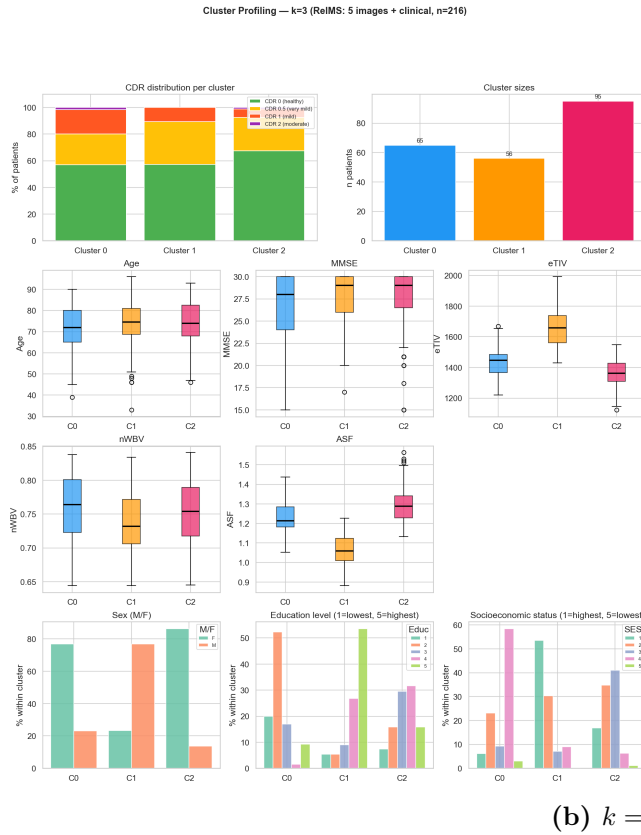
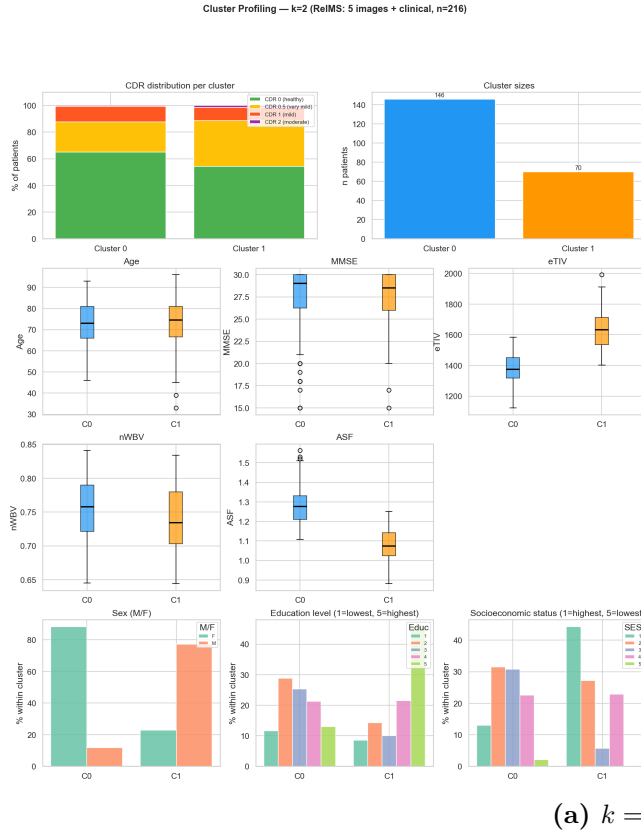


Figure 2: Cluster profiling — all 5 images + clinical. CDR distribution (top-left), cluster sizes (top-right), and boxplots for key clinical features.



Figure 3: 2D PCA scatter — all 5 images + clinical. PC1 explains 98.8% of variance; the data lies essentially on a 1D axis.

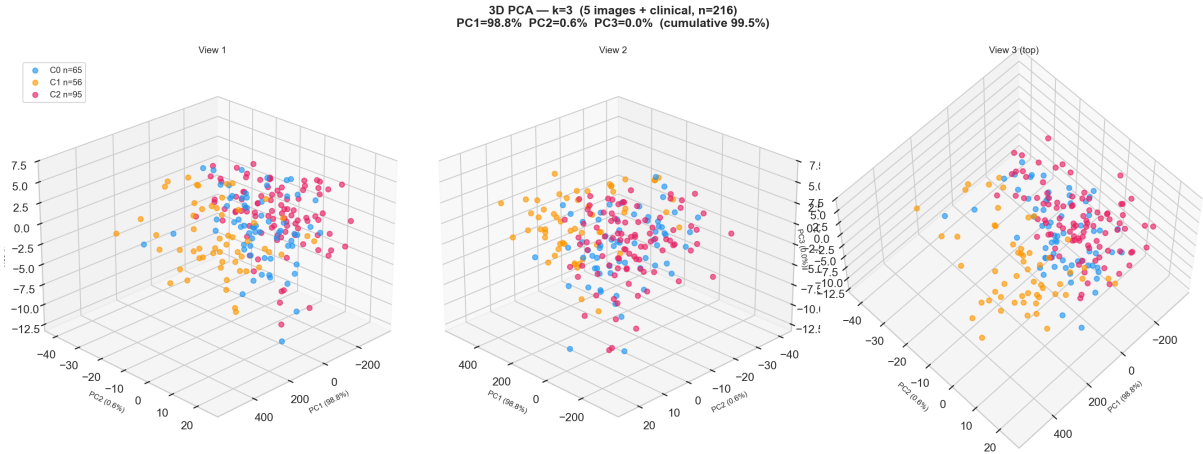


Figure 4: 3D PCA scatter — all 5 images + clinical, $k = 3$ (three viewpoints). PC1=98.8%, PC2=0.6%, PC3=0.0% (cumulative 99.5%). The third dimension carries no information; clusters overlap heavily along the single dominant axis.

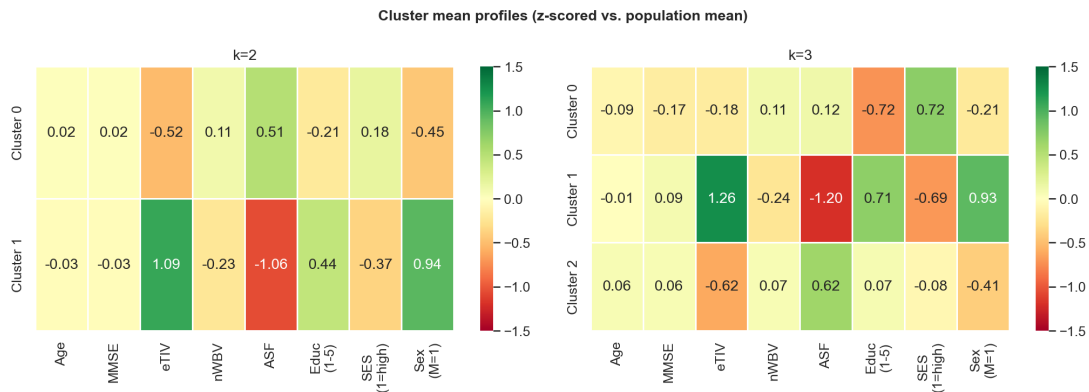


Figure 5: Cluster mean heatmap (z-scored). Each cell shows how a cluster's mean deviates from the population mean. eTIV and sex (MF) are the strongest differentiators.

5 Experiment 2: Clinical Data Only

5.1 Configuration

Feature matrix: 5 continuous clinical features (p1) + 3 ordinal/nominal features (p3).
Shape: 216×8 . Distance: RelMS.

5.2 Evaluation

Table 6: Evaluation metrics — clinical only.

k	Silhouette	ARI vs CDR	Sizes
2	0.0292	0.0245	147 / 69
3	0.0130	0.0177	78 / 62 / 76

5.3 Cluster Profiles

$k = 2$ — Almost identical to the full experiment, confirming that clinical features dominate the distance matrix:

Table 7: Cluster profiles — clinical only, $k = 2$.

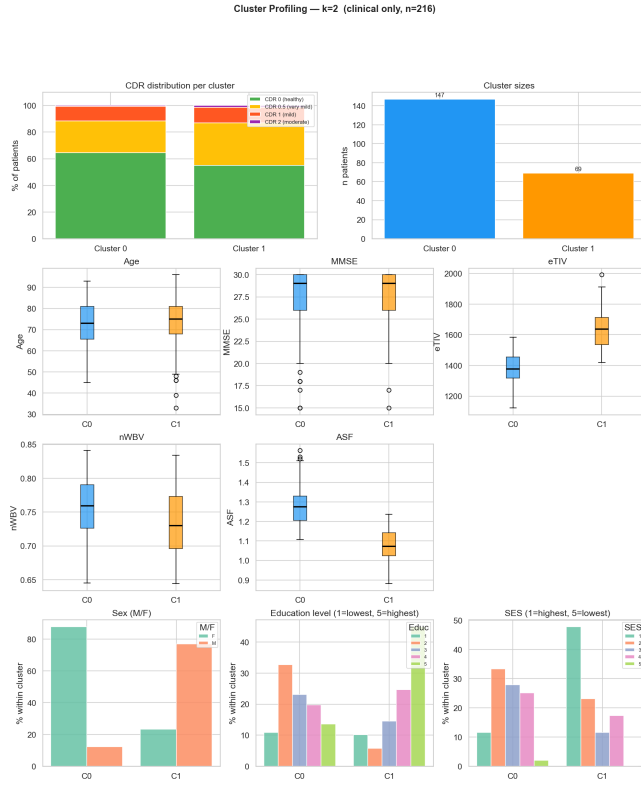
Feature	C0 ($n = 147$)	C1 ($n = 69$)
% Female	87.8%	23.2%
Age	72.3 ± 11.8	72.8 ± 13.5
MMSE	27.3 ± 3.6	27.5 ± 3.2
eTIV	1377 ± 95	1633 ± 131
Education	2.93 ± 1.23	3.88 ± 1.32
SES	2.73 ± 1.03	1.99 ± 1.14
CDR=0	64.6%	55.1%
CDR=0.5	23.8%	31.9%
CDR=1	10.9%	11.6%
CDR=2	0.7%	1.4%

$k = 3$ — Three groups emerge defined by sex, brain size, and socioeconomic factors:

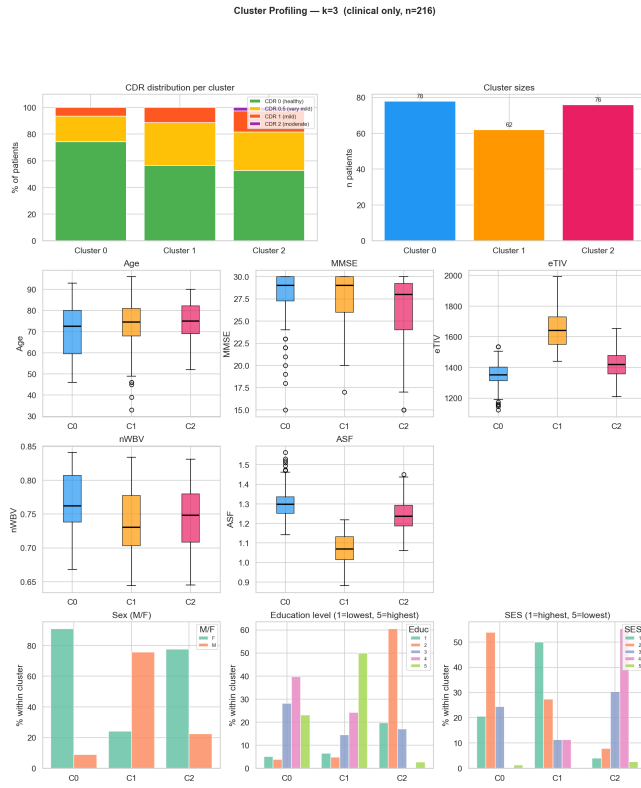
Table 8: Cluster profiles — clinical only, $k = 3$.

Feature	C0 ($n = 78$)	C1 ($n = 62$)	C2 ($n = 76$)
% Female	91.0%	24.2%	77.6%
Age	70.1 ± 13.1	72.2 ± 13.9	75.0 ± 9.4
MMSE	27.9 ± 3.2	27.6 ± 2.9	26.6 ± 3.9
eTIV	1354 ± 96	1649 ± 125	1411 ± 89
Education	3.72 ± 1.03	4.06 ± 1.20	2.05 ± 0.78
SES	2.08 ± 0.75	1.84 ± 1.03	3.45 ± 0.84
CDR=0	74.4%	56.5%	52.6%
CDR=0.5	19.2%	32.3%	28.9%
CDR=1	6.4%	11.3%	15.8%
CDR=2	0.0%	0.0%	2.6%

5.4 Visualisations



(a) $k = 2$



(b) $k = 3$

Figure 6: Cluster profiling — clinical only.



Figure 7: 2D PCA scatter — clinical only. PC1 = 99.4% of variance.

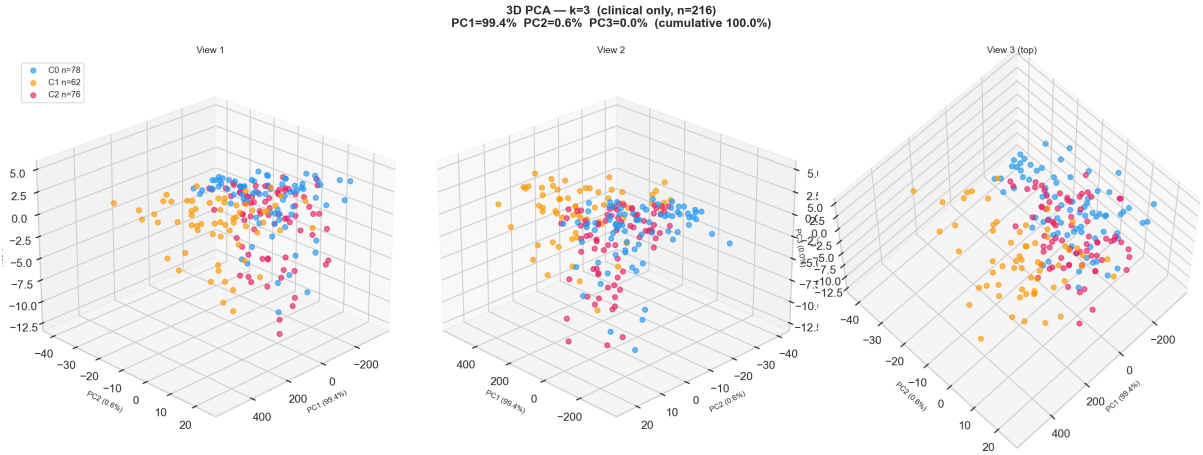


Figure 8: 3D PCA scatter — clinical only, $k = 3$ (three viewpoints). PC1=99.4%, PC2=0.6%, PC3=0.0%. The dataset is effectively 1-dimensional; no additional separation is visible in the third dimension.

6 Experiment 3: Images Only (Baseline)

6.1 Configuration

Feature matrix: five image PCA blocks concatenated (no clinical features). Shape: 216×550 . Distance: Euclidean (all features are PCA components on comparable scales; no mixed types).

6.2 Evaluation

Table 9: Evaluation metrics — images only.

k	Silhouette	ARI vs CDR	Sizes
2	0.0510	-0.0040	81 / 135
3	0.0376	-0.0205	35 / 100 / 81

The ARI vs CDR is negative for both k values: image-only clusters are *worse than random* at predicting CDR category. The slightly higher silhouette (compared to clinical experiments) reflects more uniform Euclidean distances, not meaningful cluster quality.

6.3 Image PCA Structure

Unlike clinical data, image variance is distributed across many components:

- PC1: 8.18% PC2: 4.80% PC3: 4.24% PC4: 3.09% PC5: 2.82%
- Cumulative top 5: 23.1%

The first 5 components of the combined image space explain only 23% of variance, compared to 99%+ in the clinical space. Image data is far higher-dimensional.

6.4 Cluster Profiles — PC1–PC5

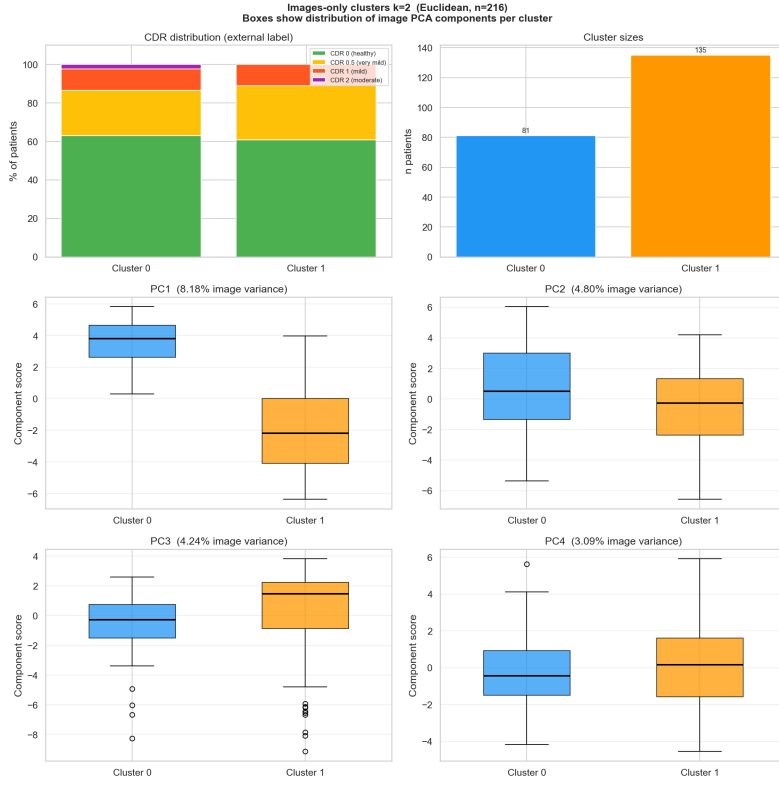
Since clinical features were not used, clusters are profiled by their image PCA components. CDR and M/F are shown as external labels only.

Table 10: Image PCA components per cluster — images only, $k = 2$. CDR and M/F are external labels not used in clustering.

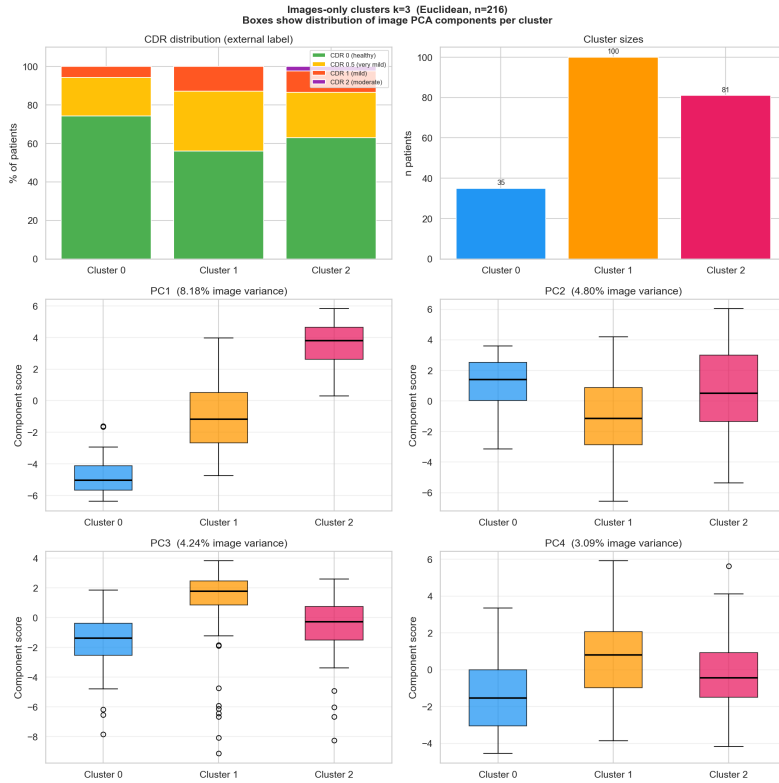
Component	C0 ($n = 81$)	C1 ($n = 135$)
PC1 (8.18%)	$+3.44 \pm 1.49$	-2.06 ± 2.46
PC2 (4.80%)	$+0.83 \pm 2.75$	-0.50 ± 2.41
PC3 (4.24%)	-0.60 ± 1.98	$+0.36 \pm 2.65$
PC4 (3.09%)	-0.21 ± 1.88	$+0.12 \pm 2.22$
PC5 (2.82%)	$+0.25 \pm 1.82$	-0.15 ± 2.10
CDR=0 (ext.)	63.0%	60.7%
% Female (ext.)	58.0%	72.6%

The split is almost entirely on PC1; PC2–PC5 differences are small relative to within-cluster standard deviations. The CDR proportions are virtually identical between clusters.

6.5 Visualisations



(a) $k = 2$



(b) $k = 3$

Figure 9: Cluster profiling — images only. Boxes show PC1–PC4 distributions per cluster. CDR (top-left) is an external label. The split is driven almost entirely by PC1.

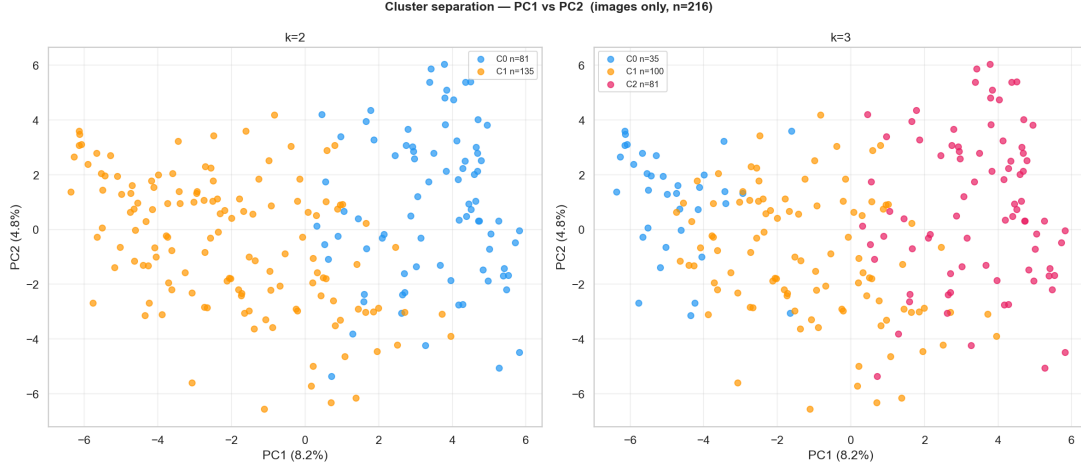


Figure 10: 2D PCA scatter — images only. Variance is more evenly distributed across components than in clinical experiments (PC1=8.2%, PC2=4.8%).

7 Experiment 4: Best Image Combination + Clinical

7.1 Configuration and Rationale

Two image types are selected: `cor` (coronal, Talairach-registered) and `gfc_sag` (sagittal, Talairach-registered). This choice is motivated by:

- Two anatomically distinct views (different planes — no slice overlap)
- Both are globally registered to Talairach space, enabling fair comparison across patients
- Avoids known methodological issues: `sbj_sag` is unregistered (native space); `masked_tra` and `gfc_tra` share the same anatomical slice (slice 90), with `masked_tra` being the skull-stripped version of `gfc_tra` (adding 45% zero HOG features)

Feature matrix: $[PCA_{cor} \mid PCA_{gfc_sag} \mid \text{clinical}]$. Shape: 216×228 . Distance: RelMS with $p_1 = [110, 110, 5]$, $p_3 = 3$.

7.2 Evaluation

Table 11: Evaluation metrics — cor+gfc_sag+clinical.

k	Silhouette	ARI vs CDR	ARI vs Exp. 1	Sizes
2	0.0269	0.0320	0.6721	149 / 67
3	0.0112	0.0161	0.5045	76 / 58 / 82

The ARI vs Experiment 1 (all 5 images + clinical) shows that using only two clean images recovers 67% of the patient groupings from the full 5-image experiment at $k = 2$, and 50% at $k = 3$.

7.3 Cluster Profiles

$k = 2$:

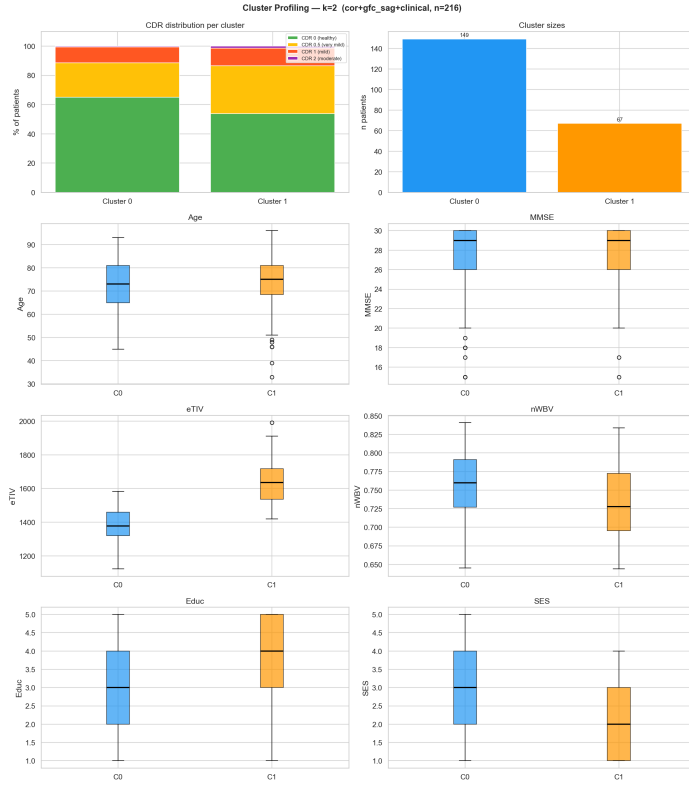
Table 12: Cluster profiles — cor+gfc_sag+clinical, $k = 2$.

Feature	C0 ($n = 149$)	C1 ($n = 67$)
% Female	86.6%	23.9%
Age	72.2 ± 11.7	72.9 ± 13.6
MMSE	27.3 ± 3.5	27.4 ± 3.2
eTIV	1379 ± 96	1635 ± 132
Education	2.93 ± 1.23	3.90 ± 1.32
SES	2.73 ± 1.03	1.96 ± 1.13
CDR=0	65.1%	53.7%
CDR=0.5	23.5%	32.8%
CDR=1	10.7%	11.9%
CDR=2	0.7%	1.5%

 $k = 3$:**Table 13:** Cluster profiles — cor+gfc_sag+clinical, $k = 3$.

Feature	C0 ($n = 76$)	C1 ($n = 58$)	C2 ($n = 82$)
% Female	75.0%	25.9%	89.0%
Age	74.4 ± 10.4	71.4 ± 14.4	71.4 ± 12.3
MMSE	26.6 ± 3.8	27.7 ± 3.0	27.8 ± 3.3
eTIV	1429 ± 94	1650 ± 129	1350 ± 97
Education	2.20 ± 1.06	4.24 ± 1.10	3.48 ± 1.02
SES	3.32 ± 1.00	1.71 ± 0.96	2.28 ± 0.81
CDR=0	52.6%	58.6%	72.0%
CDR=0.5	27.6%	31.0%	22.0%
CDR=1	18.4%	10.3%	4.9%
CDR=2	1.3%	0.0%	1.2%

7.4 Visualisations



(a) $k = 2$



(b) $k = 3$

Figure 11: Cluster profiling — cor+gfc_sag + clinical.

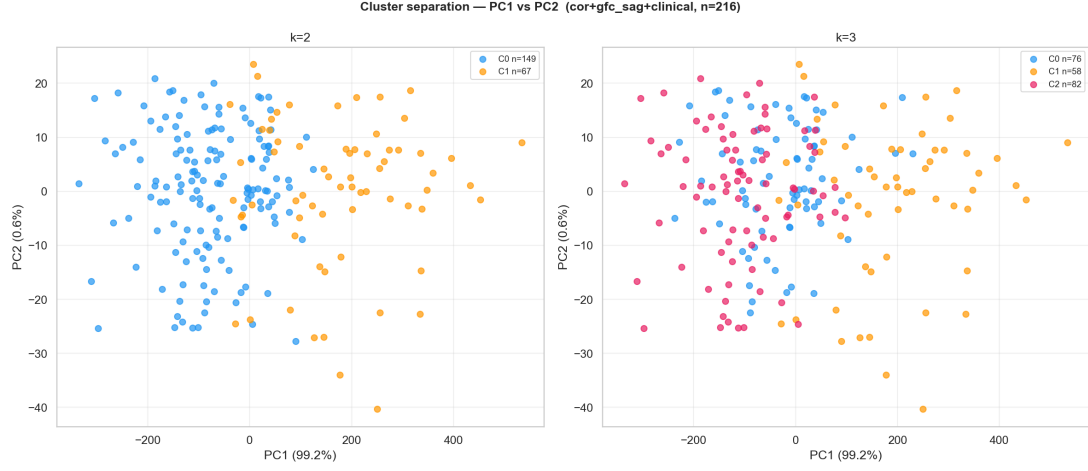


Figure 12: 2D PCA scatter — cor+gfc_sag + clinical. PC1 = 99.2%.

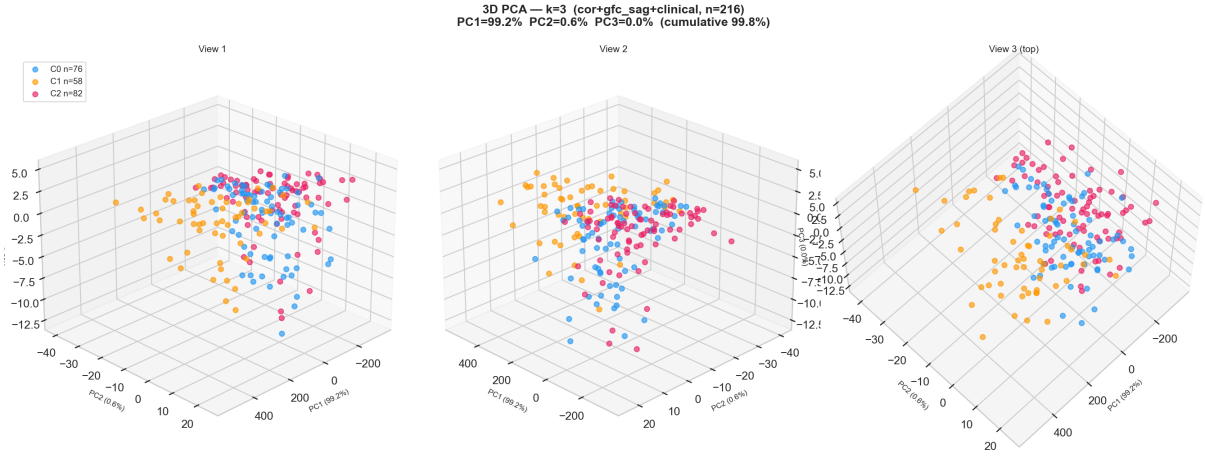


Figure 13: 3D PCA scatter — cor+gfc_sag+clinical, $k = 3$. PC1=99.2%, PC2=0.6%, PC3=0.0%. Same pattern as Experiments 1 and 2: the feature space is effectively 1-dimensional.

8 Cross-Experiment Comparison

Table 14: Summary of all four experiments. Silhouette and ARI vs CDR for $k = 2$ and $k = 3$. PC1 variance shows the effective dimensionality of each feature space.

Experiment	Silhouette		ARI vs CDR		PC1 variance	
	$k = 2$	$k = 3$	$k = 2$	$k = 3$	PC1	PC2
All 5 images + clinical	0.026	0.009	0.028	0.018	98.8%	0.6%
Clinical only	0.029	0.013	0.025	0.018	99.4%	0.6%
Images only	0.051	0.038	-0.004	-0.021	8.2%	4.8%
cor+gfc_sag+clinical	0.027	0.011	0.032	0.016	99.2%	0.6%

Key observations:

- **All silhouette scores are very low** (< 0.06): There is no strong natural cluster structure in this dataset at any tested k .
- **ARI vs CDR is near zero or negative in all experiments**: None of the discovered clusters correspond to CDR-defined disease stages.
- **Images only has the highest silhouette but the worst ARI**: More uniform Euclidean distances produce slightly tidier partitions, but those partitions have no clinical meaning.
- **Clinical data dominates all combined experiments**: The clinical-only and 5-images+clinical experiments produce nearly identical cluster assignments. The cor+gfc_sag+clinical configuration recovers 67% of the full-set assignments at $k = 2$, confirming the same dominance.
- **Feature space is effectively 1-dimensional wherever clinical data is used**: PC1 explains 98.8–99.4% of variance in all clinical configurations, driven by eTIV. The 3D PCA visualisations confirm no additional structure in PC2 or PC3.

9 Feature Separation Between Clusters

The evaluation metrics above (silhouette, ARI, NMI) answer *how good* the clusters are. This section answers a different and more informative question: **which features actually differ between the clusters, and by how much?** This is the practical output of the analysis — it tells us what each cluster *is*, in terms of the original clinical variables.

9.1 Method: Effect Sizes

For each feature we compute an **effect size** that measures how strongly cluster membership explains that feature’s values. Unlike a raw mean difference, an effect size is scale-free (0 to 1) so all features can be compared directly.

Table 15: Effect-size measures used.

Measure	Feature type	Interpretation
η^2 (eta-squared)	Continuous / ordinal	Fraction of the feature’s variance explained by the clusters
Cramér’s V	Nominal (M/F)	Strength of association between feature and cluster

The η^2 statistic is derived from the Kruskal–Wallis H test:

$$\eta^2 = \frac{H - k + 1}{n - k}$$

where k is the number of clusters and $n = 216$. The conventional interpretation is: $\eta^2 \approx 0.01$ is a *small* effect, 0.06 is *medium*, and 0.14 or above is a *large* effect. Intuitively, $\eta^2 = 0.56$ for eTIV means that 56% of the variation in brain volume is explained purely by which cluster a patient belongs to — an extremely strong separation.

9.2 Result: Which Features Drive the Clustering

Figure 14 is the central result of this section. Each cell shows the η^2 of one clinical feature (row) in one experiment (column). Darker red means the feature differs more strongly between the clusters.

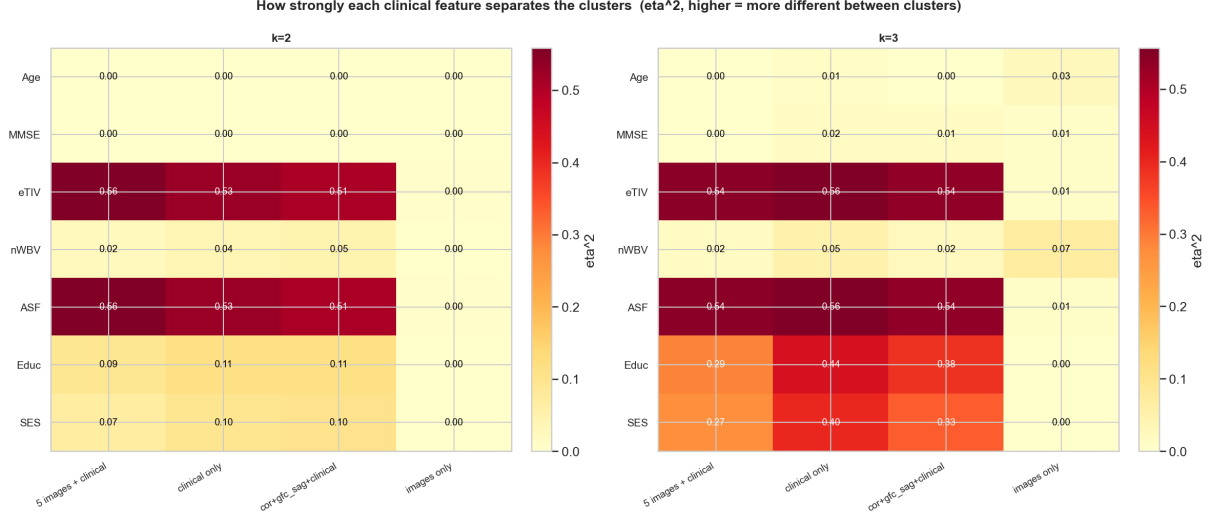


Figure 14: Feature separation (η^2) of each clinical feature across all four experiments, at $k = 2$ (left) and $k = 3$ (right). Darker = the feature differs more strongly between clusters. For the *images only* column these values are **external** (the clinical features were never used in that clustering), so they reveal whether image-derived clusters accidentally recover any clinical structure.

Table 16 gives the exact numbers, ranked, including the nominal M/F feature (Cramér’s V) and the external CDR alignment.

Table 16: Effect sizes of every feature in every experiment. M/F uses Cramér’s V; all others use η^2 . CDR is an external label (never used in clustering). Bold = large effect (≥ 0.14).

	5 img + clin		clinical only		cor+gfc_sag		images only	
Feature	k=2	k=3	k=2	k=3	k=2	k=3	k=2	k=3
M/F (sex)	.64	.56	.63	.59	.61	.55	.14	.17
eTIV	.56	.54	.53	.56	.51	.54	.00	.01
ASF	.56	.54	.53	.56	.51	.54	.00	.01
Educ	.09	.29	.12	.44	.11	.38	.00	.00
SES	.07	.27	.10	.40	.10	.33	.00	.00
nWBV	.02	.02	.04	.05	.05	.02	.00	.07
Age	.00	.00	.00	.01	.00	.00	.00	.03
MMSE	.00	.00	.00	.02	.00	.01	.00	.01
CDR (ext.)	.13	.14	.10	.17	.12	.16	.13	.13

9.3 Interpretation

Reading the heatmap and table together gives a clear and consistent story:

- **Sex and brain volume dominate every clustering.** M/F (Cramér’s $V \approx 0.55$ – 0.64), eTIV and ASF ($\eta^2 \approx 0.51$ – 0.56) are *large* effects in all three experiments that use clinical data. The clusters are, first and foremost, a split between smaller-brained females and larger-brained males.
- **ASF and eTIV are identical** because they are mathematically redundant ($\text{ASF} = 1.2109/\text{eTIV}^{1/3}$). They always have the same η^2 , which effectively double-counts brain volume in the distance metric.
- **Age and MMSE never separate the clusters** ($\eta^2 \approx 0.00$). These are the two features most directly related to cognitive decline, yet the clustering is completely blind to them. This is the clearest evidence that the clusters are demographic, not disease-related.
- **Education and SES only matter at $k = 3$.** At $k = 2$ they are small ($\eta^2 \approx 0.07$ – 0.12), but at $k = 3$ they become large ($\eta^2 \approx 0.27$ – 0.44). This tells us exactly what the third cluster is: when the algorithm is forced to find a third group, it splits patients by socioeconomic status, not by disease.
- **Images carry no clinical signal.** The entire *images only* column is near zero for every clinical feature ($\eta^2 \approx 0.00$). Even though these clusters were built from 550 image features, they fail to recover brain volume, sex, education, or any clinical variable. The HOG image features do not encode the demographic or clinical structure of the patients.
- **CDR alignment is weak everywhere.** The external CDR row stays around 0.10–0.17 (weak association). It nudges just past the “large” threshold only for clinical-only and cor+gfc_sag at $k = 3$ — the same two configurations where education/SES became strong — but never reaches a level that would support using the clusters to predict disease stage.

For completeness, the images-only clusters were also profiled by their own image PCA components. At $k = 3$, principal component 1 has $\eta^2 = 0.74$ (very large) and component 3 has $\eta^2 = 0.32$, while components 2, 4, and 5 are small. The image clustering is therefore essentially a split along a single dominant image direction (PC1), which most likely encodes global brain size and shape — the same information that eTIV already provides in the clinical data.